SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: CSA0713 **COURSE NAME: COMPUTER NETWORKS**

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 40%

Annexure A

ANALYTICAL PROBLEM SOLVING

Code of Conduct:

I C.BHUVANESH (192521108) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:C.BHUVANESH

Annexure A

ANALYTICAL PROBLEM SOLVING

1. Suppose we want to transmit the message 11100011 and protect it from errors using the CRC polynomial X3+1.

i.) Use polynomial long division to determine the message that should be transmitted.

ii) Suppose the leftmost bit of the message is inverted due to noise on the transmission link. What is the result of the receiver's CRC calculation? How does the receiver know that an error has occurred?

i) Transmitted Message

- Message = **11100011**
- Generator Polynomial = **1001** (x3+1x^3+1x3+1)
- Append 3 zeros: **11100011000**
- CRC Remainder = **100**

Transmitted Message = 11100011100

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ii) Error Detection

- Received Message (leftmost bit inverted): **01100011100**
- CRC Remainder = **110**

Since the remainder is **not 000**, the receiver detects that **an error has occurred**.

2. a) A network with bandwidth of 10 Mbps can pass only an average of 12,000 frames per minute with each frame carrying an average of 10,000 bits. What is the throughput of this network?
b) A network has a bandwidth of 3000 Hz (300 to 3300 Hz) assigned for data communications. The signal- to-noise ratio is 3162. Find the channel capacity

(a) Throughput

Given:

- Bandwidth = **10 Mbps**
- Frames = **12,000 frames/min**
- Frame size = **10,000 bits**

Calculation:

Throughput=12000×1000060=2,000,000 bps=2 Mbps\text{Throughput}=\frac{12000 \times 10000}{60} =2,000,000 \text{ bps} =2 \text{ Mbps}Throughput=6012000×10000=2,000,000 bps=2 Mbps

Answer: Throughput = 2 Mbps

(b) Channel Capacity

Given:

- Bandwidth (BBB) = **3000 Hz**
- Signal-to-Noise Ratio (SNR) = **3162**

Using Shannon's Formula:

C=Blog2(1+SNR)C = B \log_2(1 + SNR)C=Blog2(1+SNR) C=3000×log2(3163)C = 3000 \times \log_2(3163)C=3000×log2(3163) log2(3163)≈11.63\log_2(3163) \approx

11.63log2(3163)≈11.63 C=3000×11.63=34,890 bpsC = 3000 \times 11.63 = 34,890 \text{ bps}C=3000×11.63=34,890 bps

Answer: Channel Capacity ≈ 34.9 kbps

3. A multispecialty hospital uses wireless patient-monitoring devices to continuously transmit patients' heart rate and blood pressure readings to a central monitoring station. Since the data is transmitted over a wireless channel, electromagnetic interference occasionally changes one bit during transmission. To ensure that doctors receive accurate patient information without waiting for retransmission, the hospital implements the **Hamming (12,8)** error-correcting code.

Hamming (12,8) Error-Correcting Code (Short Answer)

The hospital uses **Hamming (12,8)** code to detect and correct **single-bit errors** that occur due to electromagnetic interference during wireless transmission.

- **Data bits:** 8
- **Parity bits:** 4
- **Total bits transmitted:** 12

If one bit changes during transmission:

1. The receiver checks the parity bits.
2. The error position is identified using the syndrome.
3. The incorrect bit is flipped back to its correct value.
4. The corrected patient data is delivered without retransmission.

Answer: Hamming (12,8) code detects and corrects **one-bit errors**, ensuring accurate and reliable patient-monitoring data.

4. a) A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a **/27 subnet mask** for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address **172.16.10.78** belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.

Hamming (12,8) Error-Correcting Code (Short Answer)

The hospital uses **Hamming (12,8)** code to detect and correct **single-bit errors** that occur due to electromagnetic interference during wireless transmission.

- **Data bits:** 8
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- **Total bits transmitted:** 12

If one bit changes during transmission:

1. The receiver checks the parity bits.
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Answer: Hamming (12,8) code detects and corrects **one-bit errors**, ensuring accurate and reliable patient-monitoring data.

b) A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of **/64**, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

3(b) IPv6 Addressing (Short Answer)

Given IPv6 Address:

`2001:0db8:0000:0000:0000:ff00:0042:8329`

1. Compressed Form

`2001:db8::ff00:42:8329`

2. Expanded Form

`2001:0db8:0000:0000:0000:ff00:0042:8329`

3. Network Prefix (/64)

With a **/64** prefix, the first **64 bits** (first four blocks) represent the network.

Network Prefix:

`2001:0db8:0000:0000::/64`

4. Total Host Addresses

Host bits = **128 − 64 = 64**

Host Addresses=264\text{Host Addresses} = 2^{64}Host Addresses=264 264=18,446,744,073,709,551,6162^{64} = 18,446,744,073,709,551,616264=18,446,744,073,709,551,616

Answer: 18,446,744,073,709,551,616 host addresses.

5. A university is conducting an online semester examination for **8,000 students** simultaneously. At exactly **10:00 AM**, the examination server begins transmitting encrypted question papers to all students. The university server is connected through a **10 Gbps high-speed network**, while students access the examination portal using different internet connections such as **fiber broadband (100 Mbps), home Wi-Fi (20 Mbps), and mobile networks (5–10 Mbps)**. During the first few minutes of the examination, many students with slower internet connections report incomplete downloads, repeated retransmissions, delayed acknowledgments, and temporary freezing of the examination portal. Network administrators observe that the server continues transmitting data at a high rate even though several student devices are unable to receive and process packets at the same speed. This results in receiver buffer overflow, unnecessary retransmissions, increased network traffic, and delays in delivering the question papers. **Analyze** the above scenario and identify the primary networking problem. Explain how the mismatch between the sender's transmission speed and the receiver's processing capability affects communication.

Primary Networking Problem: Flow Control

Explanation:

The university server transmits data at **10 Gbps**, while many students have slower internet connections (**100 Mbps, 20 Mbps, or 5–10 Mbps**). Because the sender transmits faster than some receivers can process the data, the **receiver's buffer overflows**, causing **packet loss, retransmissions, delayed acknowledgments, and slow downloads**.

Effect on Communication:

The speed mismatch causes network congestion and reduces communication efficiency. Flow control mechanisms ensure that the sender transmits data only at a rate the receiver can handle, preventing buffer overflow and ensuring reliable delivery of the examination papers.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation